

Prob 1 (LO01) (20 pts, 5 pts each). Find the phasor form expression of the following voltage sources.

1. $x_a(t) = 10 \sin(\omega t + \pi/12)$.
2. $x_b(t) = 15 \cos(\omega t + \pi/12)$.
3. $x_c(t) = -12 \sin(\omega t - \pi/12)$.
4. $x_d(t) = -8 \cos(\omega t - \pi/12)$.

Prob 2 (LO01) (10 pts, 5 pts each). Consider the signals of the previous problem. Find the following using the phasors technique:

1. $x_a(t) + x_b(t)$.
2. $x_b(t) - x_c(t)$.

Prob 3 (LO02) (20 pts, 5 pts each). Consider the following DT signal

$$x[n] = \begin{cases} \cos(\frac{\pi}{6}n), & 0 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Plot the signal and its different variations using the plotting format shown in the class notes:

1. $x[n]$.
2. $x[n - 2]$.
3. $x[-n]$. (We did not discuss this in class. Try to figure it out using the flip concept discussed in LO 5.)
4. $x[2 - n]$. (We did not discuss this in class. Try to figure it out.)

Prob 4 (LO03) (20 pts, 5 pts each). Determine whether each of the following signals is periodic. For those are periodic, determine their fundamental period.

1. $x_a[n] = e^{j(\pi n/8)}$.
2. $x_b[n] = e^{j(7\pi n/9)}$.
3. $x_c[n] = ne^{j(\pi n/8)}$.
4. $x_d[n] = e^{j(\pi n/\sqrt{8})}$.

Prob 5 (LO04) (40 pts total, 2 pts for each). Determine if the each of the systems below is (1) linear, (2) time-invariant, (3) causal, and (4) BIBO stable. (No procedure is needed if you know how to perform them.)

1. $T_a(x[n]) = nx[n]$.
2. $T_b(x[n]) = ax[n] + c$, where a and c are constants.
3. $T_c(x[n]) = x[n + 1] + x[n - 1]$.
4. $T_d(x[n]) = x[n]x[n + 1]$.
5. $T_e(x[n]) = e^{x[n-1]}$.

Prob 1. (LO05) (20 pts total). Assume $x[n] = a^n u[n]$. Calculate the convolution of $y[n] = x[n] * x[n]$ using the flip, slide, and sum method.